

SQL Fundamentals

SELECT & Filtering

SELECTING & SORTING

TABLE employees

id	first-name	last-name	department	salary	hire-date	city
1	Alice	Green	IT	10000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	Mokoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72 000	2017-09-30	Johannesburg

Q1) SELECT *
FROM employees;

Q2) SELECT DISTINCT department
FROM employees;

TABLE employees

department
IT
HR
FINANCE
Marketing
IT

Q3) SELECT first-name,
last-name
FROM employees
ORDER BY salary DESC;

TABLE employees

first-name	last-name	salary
EVA	Naidoo	72 000
Alice	Green	70 000
Cathy	Zulu	65 000
David	Mokoena	50 000
Brian	Lee	45 000

Q4) SELECT id,
first_name,
last_name,
salary
FROM employees
ORDER BY salary DESC
LIMIT 3;



TABLE employees

id	first_name	last_name	salary
5	Eva	Maidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

Q5) SELECT first_name,
last_name,
department
FROM employees
WHERE department = 'IT';



TABLE employees

first_name	last_name	department
Alice	Green	IT
Eva	Naidoo	IT

Q6) SELECT *
FROM employees
WHERE department = 'Finance', AND salary > 60 000;

TABLE employees

id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65 000



Q7) SELECT id,
first_name,
last_name,
department
FROM employees

WHERE department = 'HR', OR department = 'Marketing';

Q7)

~~TABLE employees~~

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

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TABLE employees

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

Q8) SELECT id,
first-name,
last-name,
department

FROM employees

WHERE department != 'IT';

TABLE employees

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

Q9) SELECT id,
first-name,
last-name,
department

FROM employees

WHERE department IN 'IT', 'HR', 'FINANCE';

TABLE employees

id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance

Q10) SELECT *

FROM employees

WHERE department = 'IT' AND salary > 65 000, AND city = 'Johannesburg';

Table employees

Q10)

id	first-name	last-name	department	salary	hire-date	city
1	Alice	Green	IT	70 006		Johannesburg